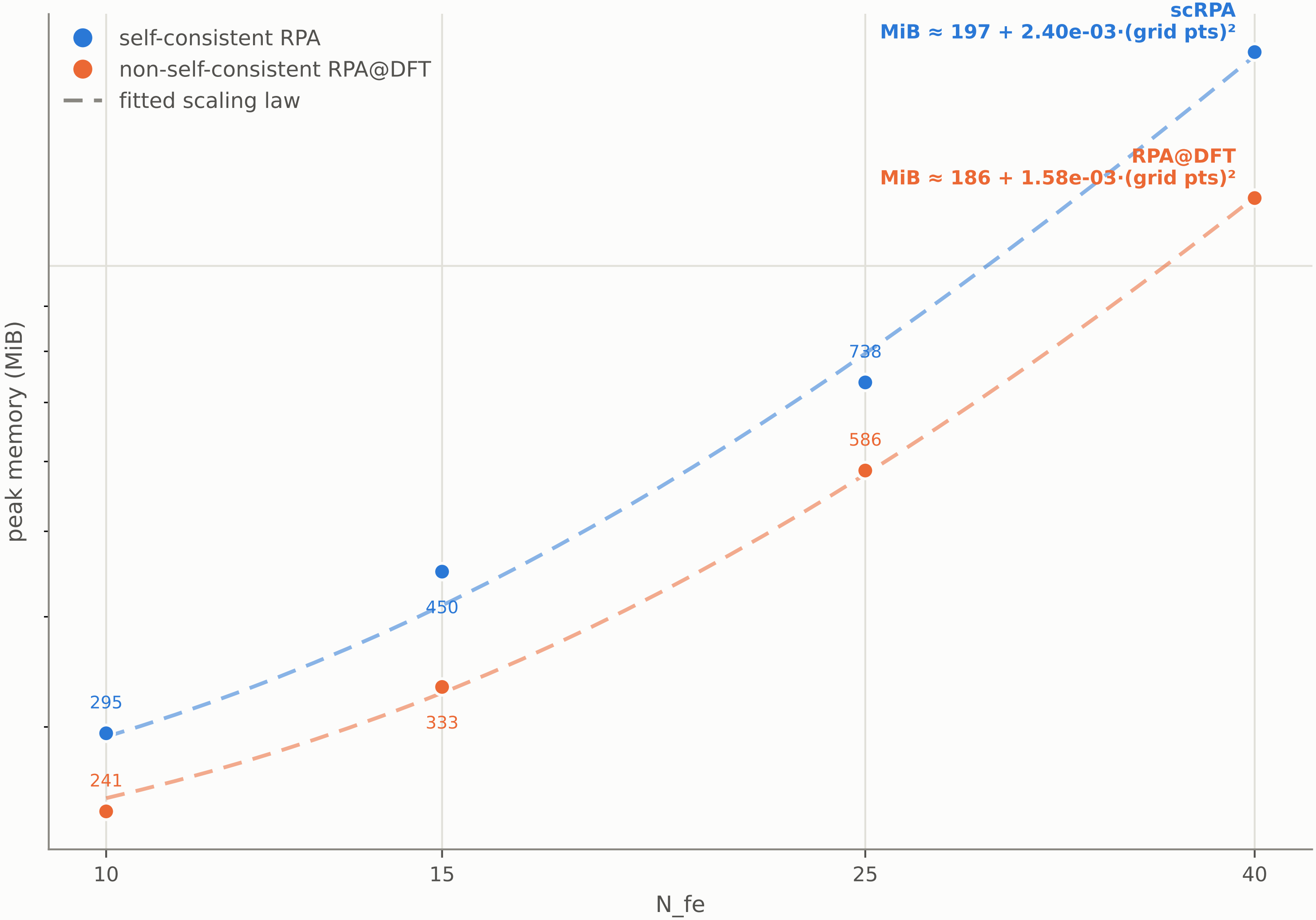
RPA scaling report

Phase 2: N_fe / Lmax sweeps -- Phase 3: quadrature order q

Mercury (Z=80), all-electron, corrected SPARC-atomSFE, full SCF (outer loop converged). Fixed: p=20, q=45, domain=13 Bohr, omega=4, polynomial mesh (concentration 2). Channels = Lmax + 4 (Hg has 4f occupied, l_occ_max = 3).

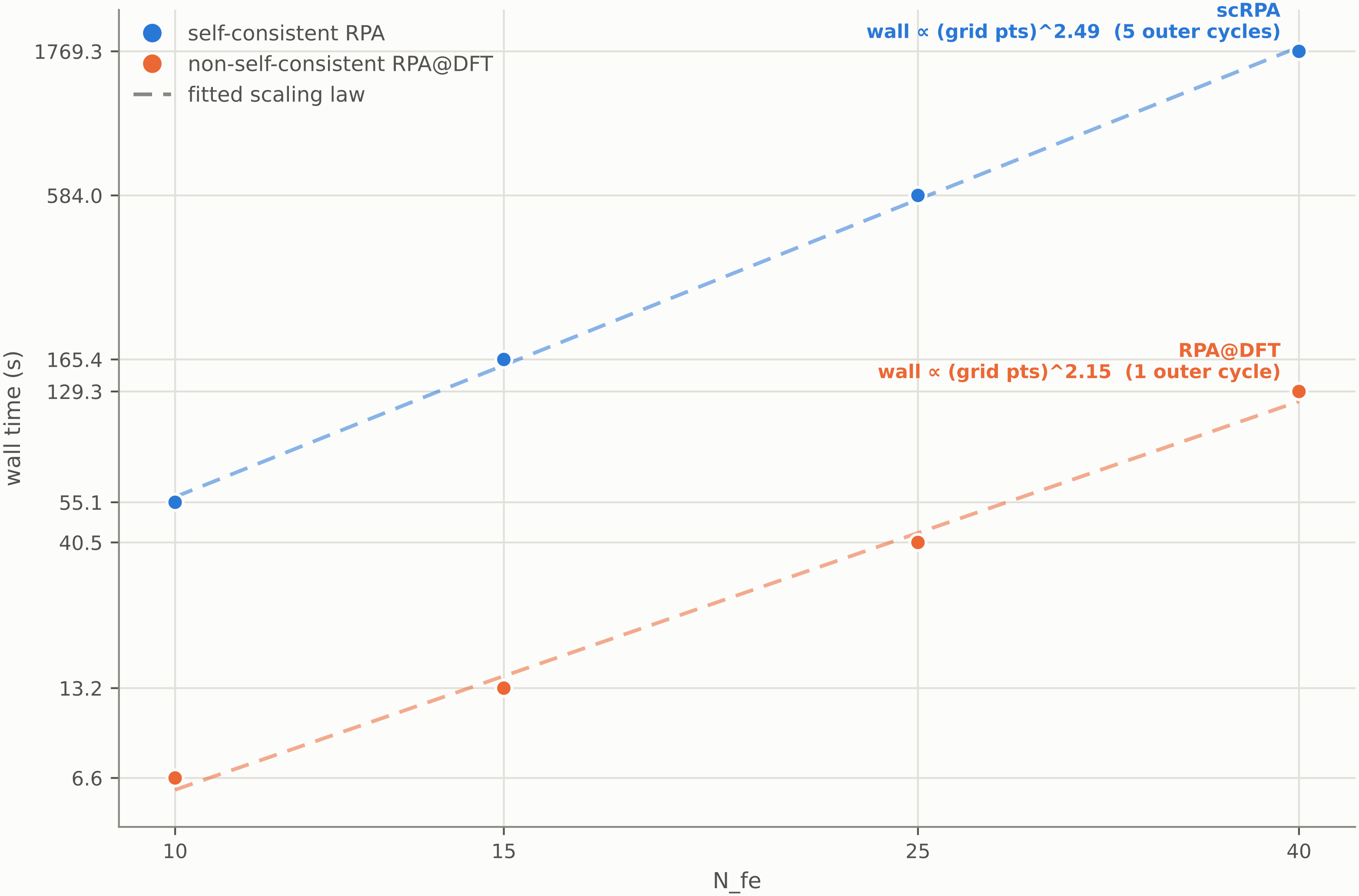
Contents

Memory vs N_fe

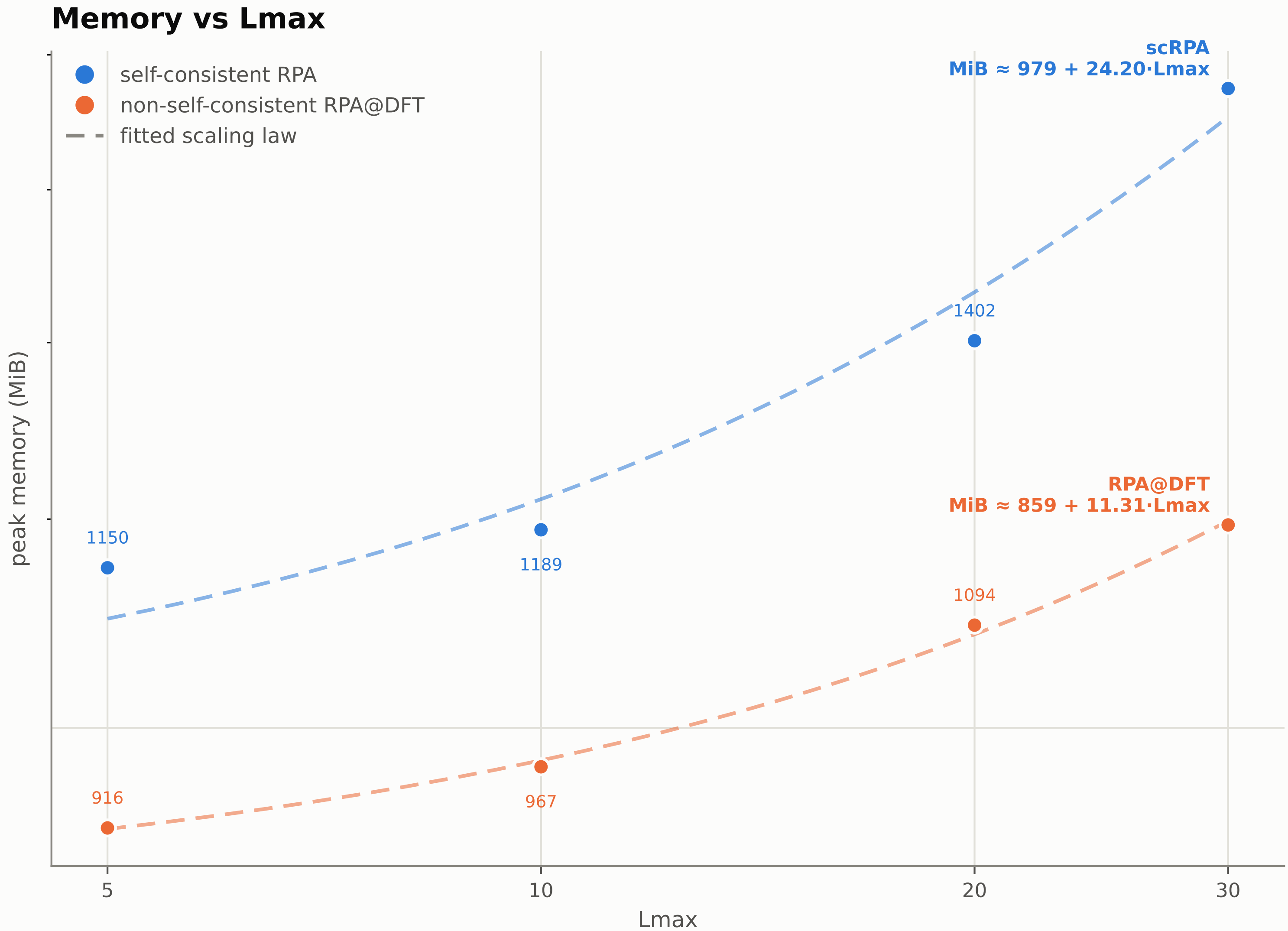


Lmax held at 30. x-axis is N_fe; the fit is against grid points (= N_fe x p - 1, the actual FE degrees of freedom -- p=20 here), since that is what the RPA kernel matrices scale with, not the quadrature order. N_fe -> grid points: 10→199, 15→299, 25→499, 40→799.

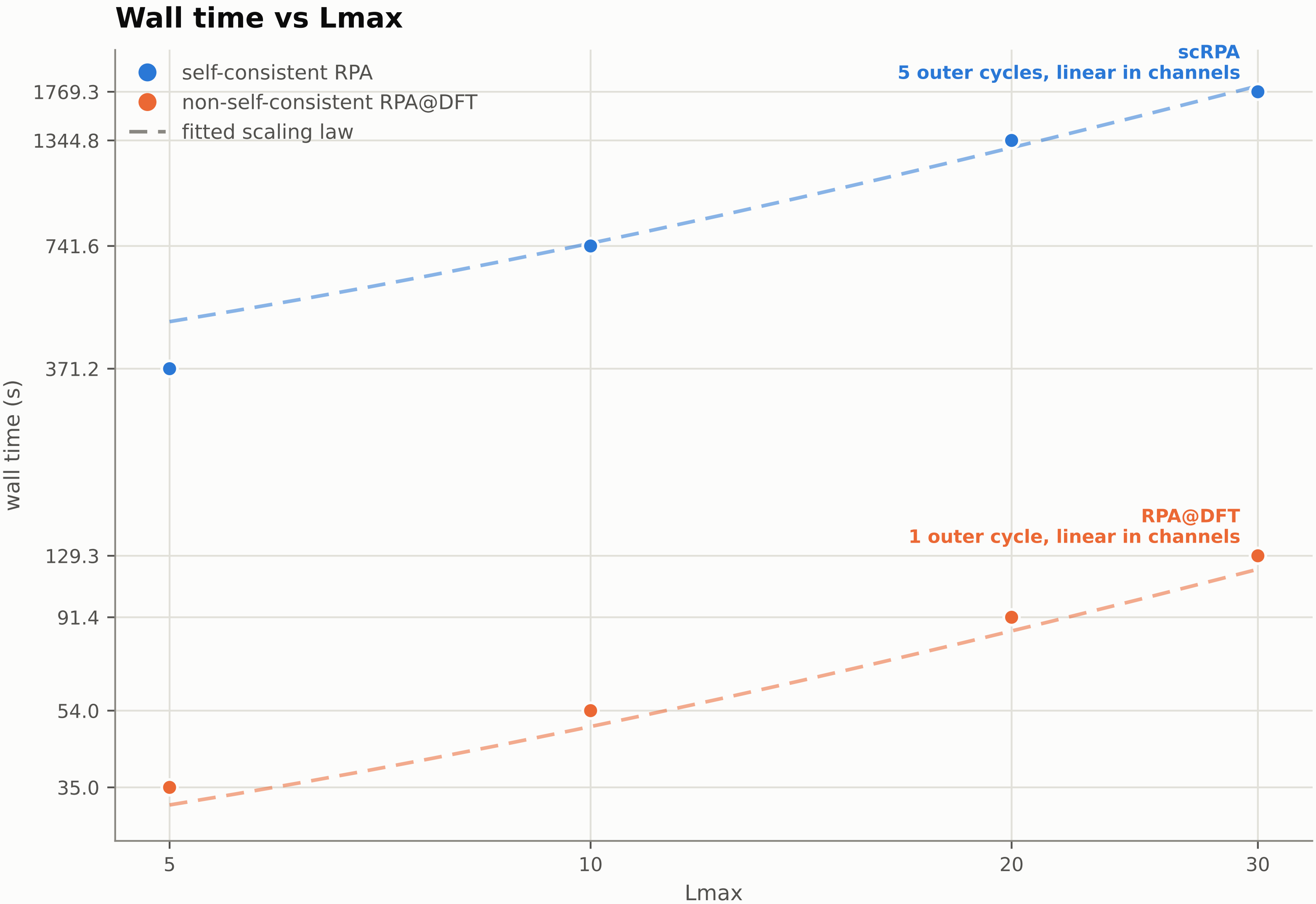
Wall time vs N_fe



Lmax held at 30. x-axis is N_fe; the fit is against grid points (= N_fe x p - 1, the actual FE degrees of freedom -- p=20 here), since that is what the RPA kernel matrices scale with, not the quadrature order. N_fe -> grid points: 10→199, 15→299, 25→499, 40→799.



N_fe held at 40 (grid points = 799), so only the channel count (Lmax + 4) moves. x-axis is Lmax.



N_fe held at 40 (grid points = 799), so only the channel count (Lmax + 4) moves. x-axis is Lmax.

N_fe sweep, numeric -- self-consistent RPA

Mercury (Z=80), all-electron, corrected SPARC-atomSFE, full SCF (outer loop converged). Fixed: p=20, q=45, domain=13 Bohr, omega=4, polynomial mesh (concentration 2). Channels = Lmax + 4 (Hg has 4f occupied, l_occ_max = 3).

N_fe	grid points	outer	mem (MiB)	wall (s)	s/channel/outer	mem exp	time exp
10	199	5	295.1	55.1	0.324	**-**	**-**
15	299	5	450.0	165.4	0.973	**1.04**	**2.7**
25	499	5	737.7	584.0	3.44	**0.965**	**2.46**
40	799	5	1747.9	1769.3	10.4	**1.83**	**2.35**

N_fe sweep, numeric -- non-self-consistent RPA@DFT

Mercury (Z=80), all-electron, corrected SPARC-atomSFE, full SCF (outer loop converged). Fixed: p=20, q=45, domain=13 Bohr, omega=4, polynomial mesh (concentration 2). Channels = Lmax + 4 (Hg has 4f occupied, l_occ_max = 3).

N_fe	grid points	outer	mem (MiB)	wall (s)	s/channel/outer	mem exp	time exp
10	199	1	240.6	6.6	0.194	**-**	**-**
15	299	1	333.0	13.2	0.388	**0.798**	**1.7**
25	499	1	586.0	40.5	1.19	**1.1**	**2.19**
40	799	1	1193.9	129.3	3.8	**1.51**	**2.47**

Lmax sweep, numeric -- self-consistent RPA

Mercury (Z=80), all-electron, corrected SPARC-atomSFE, full SCF (outer loop converged). Fixed: p=20, q=45, domain=13 Bohr, omega=4, polynomial mesh (concentration 2). Channels = Lmax + 4 (Hg has 4f occupied, l_occ_max = 3).

Lmax	channels	outer	mem (MiB)	wall (s)	s/channel/outer	mem exp	time exp
5	9	4	1150.0	371.2	10.3	**-**	**-**
10	14	5	1188.9	741.6	10.6	**0.0751**	**1.57**
20	24	5	1402.3	1344.8	11.2	**0.306**	**1.1**
30	34	5	1747.9	1769.3	10.4	**0.633**	**0.788**

Lmax sweep, numeric -- non-self-consistent RPA@DFT

Mercury (Z=80), all-electron, corrected SPARC-atomSFE, full SCF (outer loop converged). Fixed: p=20, q=45, domain=13 Bohr, omega=4, polynomial mesh (concentration 2). Channels = Lmax + 4 (Hg has 4f occupied, l_occ_max = 3).

Lmax	channels	outer	mem (MiB)	wall (s)	s/channel/outer	mem exp	time exp
5	9	1	916.3	35.0	3.89	**-**	**-**
10	14	1	966.5	54.0	3.86	**0.121**	**0.979**
20	24	1	1093.9	91.4	3.81	**0.23**	**0.977**
30	34	1	1193.9	129.3	3.8	**0.251**	**0.995**

Total-energy convergence, N_fe sweep -- self-consistent RPA

Mercury (Z=80), all-electron, corrected SPARC-atomSFE, full SCF (outer loop converged). Fixed: p=20, q=45, domain=13 Bohr, omega=4, polynomial mesh (concentration 2). Channels = Lmax + 4 (Hg has 4f occupied, l_occ_max = 3).

N_fe	Lmax	E (Ha)	dE vs previous (Ha)
10	30	-18413.639227975	**-**
15	30	-18413.641135948	**-1.91e-03**
25	30	-18413.641200805	**-6.49e-05**
40	30	-18413.641206868	**-6.06e-06**

`dE vs previous` is this row's energy minus the row above it, within the same sweep (N_fe sweep at Lmax=30; Lmax sweep at N_fe=40) -- not a comparison against any predicted or extrapolated converged value.

Total-energy convergence, Lmax sweep -- self-consistent RPA

Mercury (Z=80), all-electron, corrected SPARC-atomSFE, full SCF (outer loop converged). Fixed: p=20, q=45, domain=13 Bohr, omega=4, polynomial mesh (concentration 2). Channels = Lmax + 4 (Hg has 4f occupied, l_occ_max = 3).

N_fe	Lmax	E (Ha)	dE vs previous (Ha)
40	5	-18413.171562837	**-**
40	10	-18413.619296343	**-4.48e-01**
40	20	-18413.640585551	**-2.13e-02**
40	30	-18413.641206868	**-6.21e-04**

`dE vs previous` is this row's energy minus the row above it, within the same sweep (N_fe sweep at Lmax=30; Lmax sweep at N_fe=40) -- not a comparison against any predicted or extrapolated converged value.

Total-energy convergence, N_fe sweep -- non-self-consistent RPA@DFT

Mercury (Z=80), all-electron, corrected SPARC-atomSFE, full SCF (outer loop converged). Fixed: p=20, q=45, domain=13 Bohr, omega=4, polynomial mesh (concentration 2). Channels = Lmax + 4 (Hg has 4f occupied, l_occ_max = 3).

N_fe	Lmax	E (Ha)	dE vs previous (Ha)
10	30	-18413.631984602	**-**
15	30	-18413.631983599	**+1.00e-06**
25	30	-18413.631981613	**+1.99e-06**
40	30	-18413.631969526	**+1.21e-05**

`dE vs previous` is this row's energy minus the row above it, within the same sweep (N_fe sweep at Lmax=30; Lmax sweep at N_fe=40) -- not a comparison against any predicted or extrapolated converged value.

Total-energy convergence, Lmax sweep -- non-self-consistent RPA@DFT

Mercury (Z=80), all-electron, corrected SPARC-atomSFE, full SCF (outer loop converged). Fixed: p=20, q=45, domain=13 Bohr, omega=4, polynomial mesh (concentration 2). Channels = Lmax + 4 (Hg has 4f occupied, l_occ_max = 3).

N_fe	Lmax	E (Ha)	dE vs previous (Ha)
40	5	-18413.162880405	**-**
40	10	-18413.610108664	**-4.47e-01**
40	20	-18413.631348768	**-2.12e-02**
40	30	-18413.631969526	**-6.21e-04**

`dE vs previous` is this row's energy minus the row above it, within the same sweep (N_fe sweep at Lmax=30; Lmax sweep at N_fe=40) -- not a comparison against any predicted or extrapolated converged value.

Projections

Mercury (Z=80), all-electron, corrected SPARC-atomSFE, full SCF (outer loop converged). Fixed: p=20, q=45, domain=13 Bohr, omega=4, polynomial mesh (concentration 2). Channels = Lmax + 4 (Hg has 4f occupied, l_occ_max = 3).

mode	N_fe	Lmax	grid points	channels	outer	mem low (GiB)	mem high (GiB)	wall (h)
scRPA	150	80	2999	84	5	**22.5**	**37.9**	**33.8**
scRPA	200	100	3999	104	5	**39.4**	**79.2**	**85.6**
RPA@DFT	150	80	2999	84	1	**14.6**	**21.9**	**1.4**
RPA@DFT	200	100	3999	104	1	**25.7**	**44.2**	**3.2**

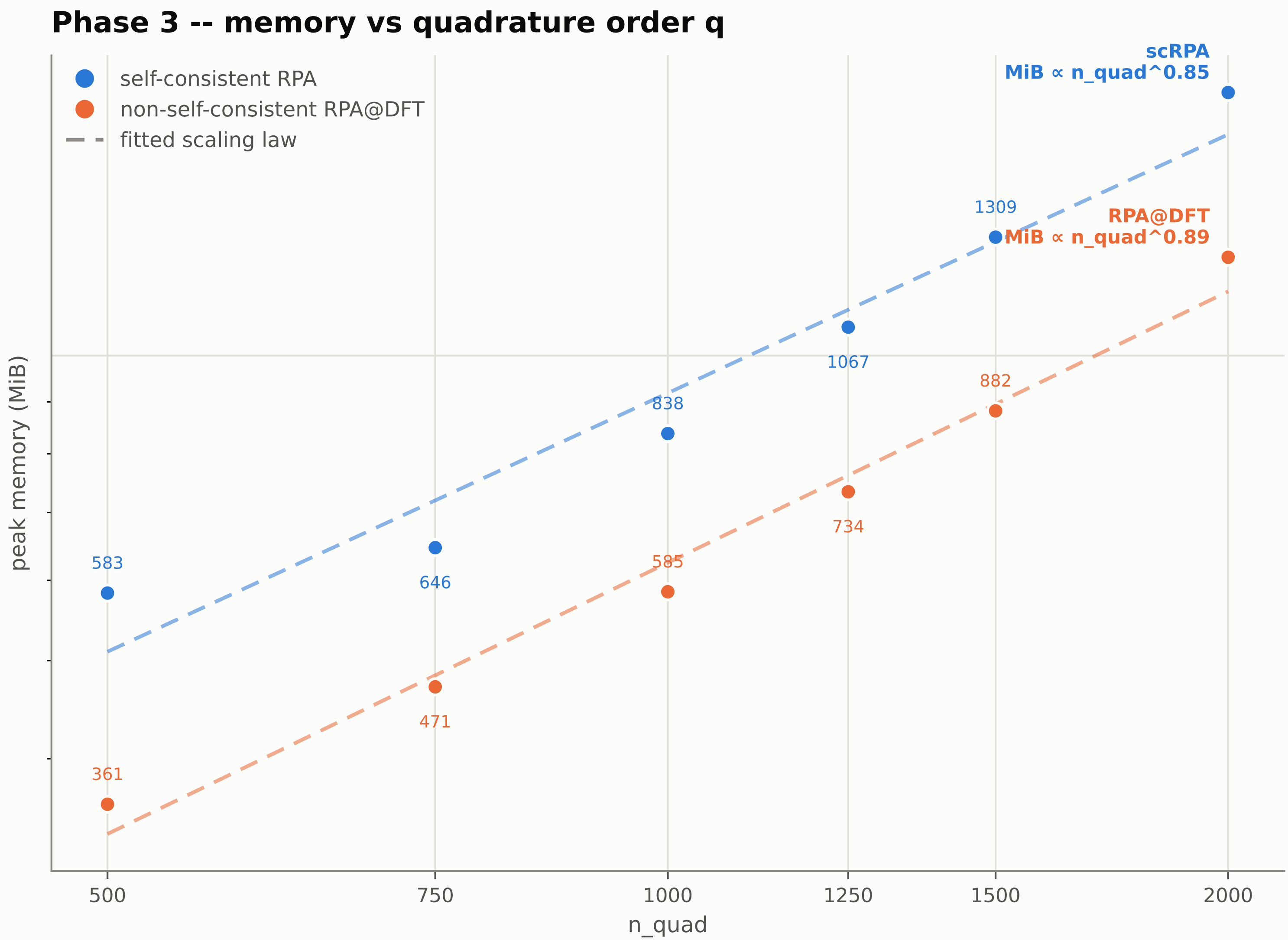
`mem low` treats the Lmax term as additive and grid-point-independent; `mem high` lets it scale with the grid-points-squared term. Only one grid-point count was sampled in the angular sweep, so which is correct is undetermined -- the true value lies between them. This is the largest uncertainty in this report. Wall time needs no such caveat: it is the measured outer-cycle count (scRPA: 5, from the N_fe=40/Lmax=30 anchor; RPA@DFT: 1, by construction) times the fitted per-channel cost.

Self-consistent / non-self-consistent cost ratio

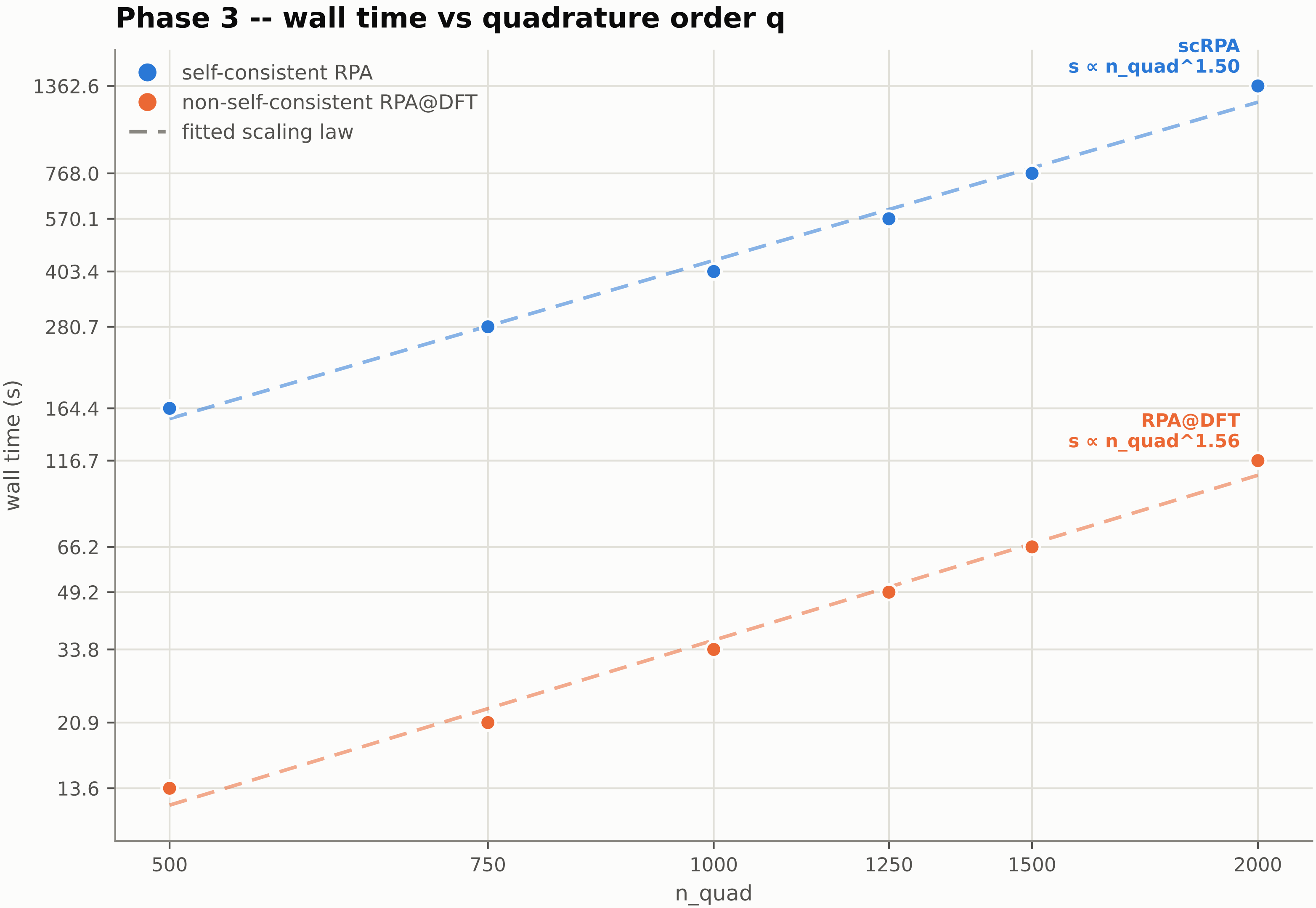
Mercury (Z=80), all-electron, corrected SPARC-atomSFE, full SCF (outer loop converged). Fixed: p=20, q=45, domain=13 Bohr, omega=4, polynomial mesh (concentration 2). Channels = Lmax + 4 (Hg has 4f occupied, l_occ_max = 3).

N_fe	Lmax	grid points	scRPA (s)	RPA@DFT (s)	ratio	scRPA MiB	RPA@DFT MiB	mem ratio	kind
10	30	199	55.1	6.6	**8.34x**	295.1	240.6	**1.23x**	measured
15	30	299	165.4	13.2	**12.53x**	450.0	333.0	**1.35x**	measured
25	30	499	584.0	40.5	**14.43x**	737.7	586.0	**1.26x**	measured
40	5	799	371.2	35.0	**10.60x**	1150.0	916.3	**1.26x**	measured
40	10	799	741.6	54.0	**13.74x**	1188.9	966.5	**1.23x**	measured
40	20	799	1344.8	91.4	**14.72x**	1402.3	1093.9	**1.28x**	measured
40	30	799	1769.3	129.3	**13.69x**	1747.9	1193.9	**1.46x**	measured
150	80	2999	121618.0	5081.5	**23.93x**	23013.4	14979.6	**1.54x**	projected
200	100	3999	308232.0	11677.9	**26.39x**	40308.5	26276.1	**1.53x**	projected

Ratio is scRPA / RPA@DFT wall time, using each mode's own measured outer-cycle count -- no per-iteration normalisation is applied since neither number was ever per-iteration. The ratio grows with grid size because scRPA's per-channel cost has the steeper exponent in grid points (2.49 vs 2.15). Shaded rows are projected.



x-axis is n_quad = N_fe x q (N_fe=25 fixed). scRPA: 4 outer cycles at every q; RPA@DFT: 1 (no outer loop).



x-axis is n_quad = N_fe x q (N_fe=25 fixed). scRPA: 4 outer cycles at every q; RPA@DFT: 1 (no outer loop).

Phase 3 -- numeric data

Mercury (Z=80), all-electron, full SCF, N_fe=25, p=20, Lmax=30, omega=8, dense basis order = p. n_quad = N_fe x q grows through q at FIXED N_fe here -- a different operation from phase 2's N_fe sweep, so the exponents below are not comparable to pages 1-2.

mode	q	n_quad	outer	mem (MiB)	wall (s)	E (Ha)
RPA@DFT	20	500	1	360.7	13.6	-18414.910538018
RPA@DFT	30	750	1	471.0	20.9	-18414.910536919
RPA@DFT	40	1000	1	584.6	33.8	-18414.910537648
RPA@DFT	50	1250	1	733.9	49.2	-18414.910538091
RPA@DFT	60	1500	1	882.0	66.2	-18414.910537448
RPA@DFT	80	2000	1	1250.6	116.7	-18414.910535844
scRPA	20	500	4	582.9	164.4	-18414.920617675
scRPA	30	750	4	646.2	280.7	-18414.920617674
scRPA	40	1000	4	837.6	403.4	-18414.920617679
scRPA	50	1250	4	1066.8	570.1	-18414.920617671
scRPA	60	1500	4	1308.9	768.0	-18414.920617671
scRPA	80	2000	4	1818.5	1362.6	-18414.920617674

Energy spread across the whole q sweep: scRPA 8.31e-09 Ha, RPA@DFT 2.25e-06 Ha -- q is converged at q=20 for both modes at this grid.